



eMERGE Data to Discovery: The i2b2-AnVIL Clinical Bridge

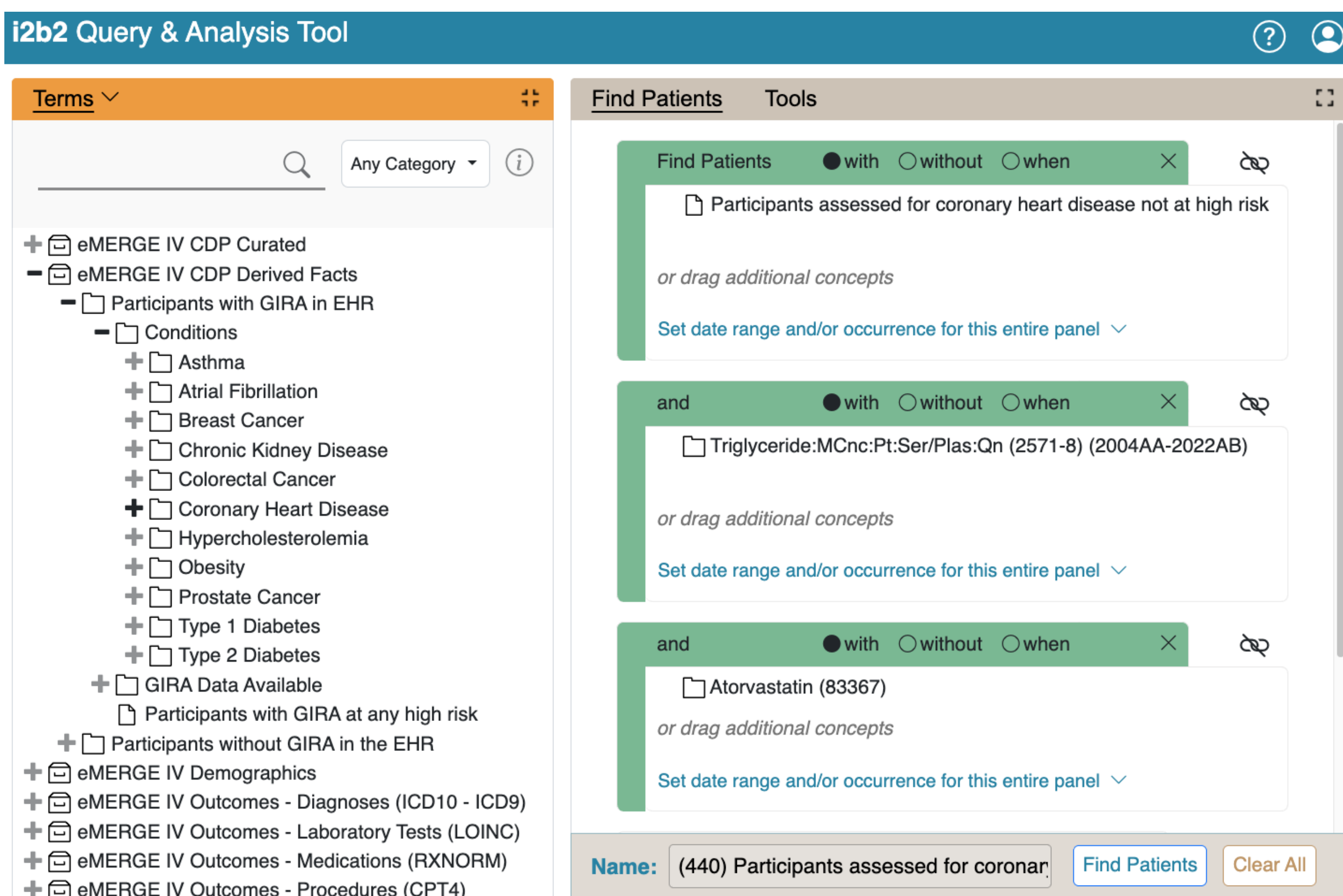
Jeffrey G. Klann, PhD^{1,2,3}; Victor M. Castro, MS²; Taowei David Wang, PhD^{2,3}; Kavishwar B. Waghlikar, MBBS, PhD^{1,2,3}; Shawn N. Murphy, MD, PhD⁴

1. Harvard Medical School 2. Mass General Brigham 3. Massachusetts General Hospital 4. University of Washington



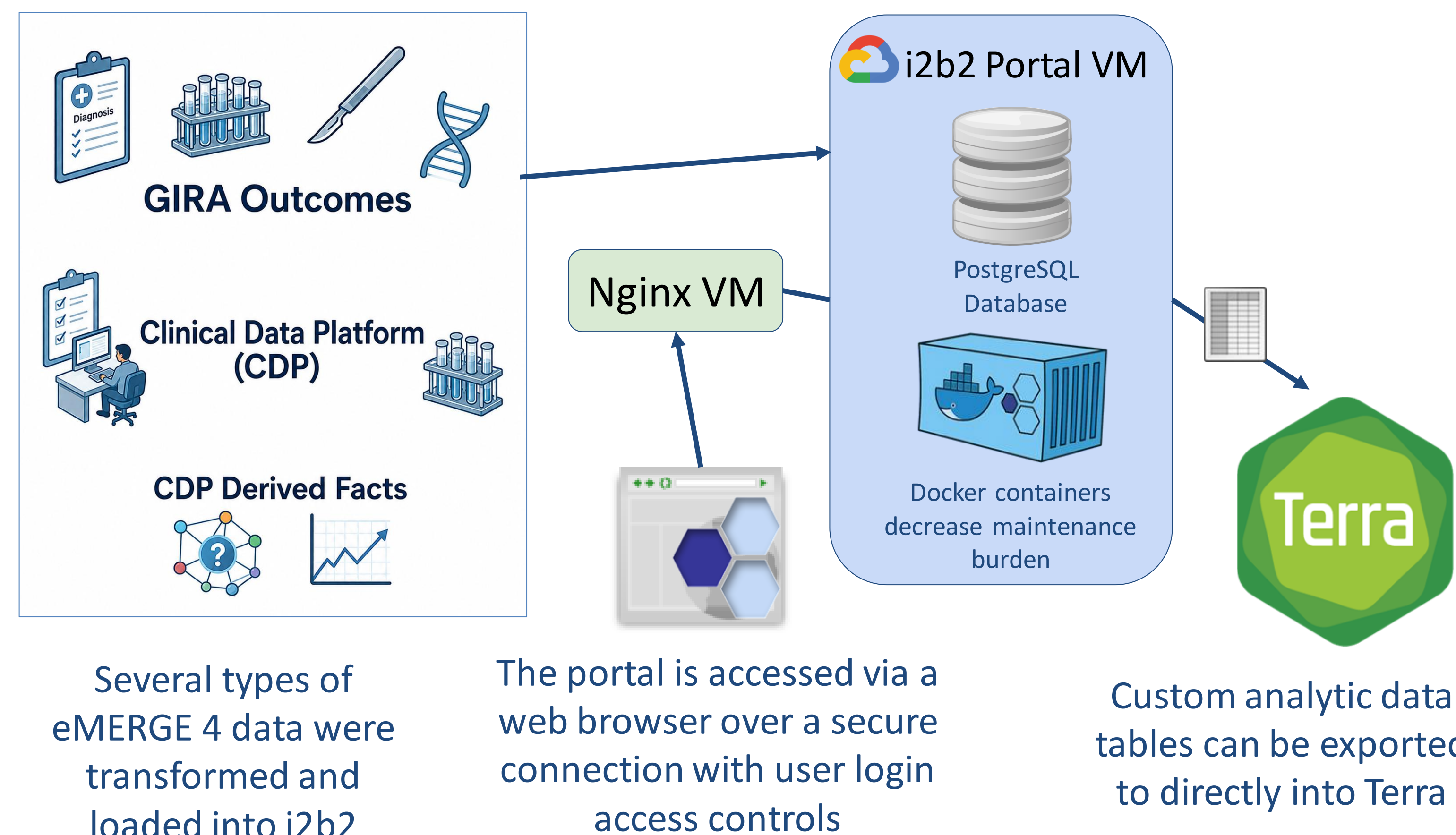
eMerge Clinical Data Platform Portal

- The AnVIL Clinical Resource (ACR) aims to bring clinical and phenotypic data sets and tools to the AnVIL platform, for use alongside genomic data for advanced analytics.
- Informatics for Integrating Biology and the Bedside (i2b2)** is a widely used, open-source clinical data warehousing and analytics platform
 - Supports any subject-oriented data, creating a pathway for importing study-specific formats.
 - Provides graphical, self-service cohort exploration and export tools.
- We developed an **Electronic Medical Records and Genomics (eMERGE)** data research portal and integrated it with AnVIL.
- This brings eMERGE 4 clinical data—EHR diagnoses, medications, procedures, labs, survey responses, and genomic indicators such as polygenic risk scores—into the i2b2 platform.
- Portal Design:** i2b2 runs on **Google Cloud Platform (GCP)** using containerized **Docker** images on Linux **virtual machines (VMs)** inside a **Virtual Private Cloud (VPC)**. Admin access via bastion host; user traffic through a **Nginx** reverse proxy.
- The portal has been in development since April 2025; beta tested at the end of 2025; and launched broadly to eMERGE investigators in August 2026.



The portal interface used for a multimodal query

Portal Design



Clinical Data Platform REDCap Data



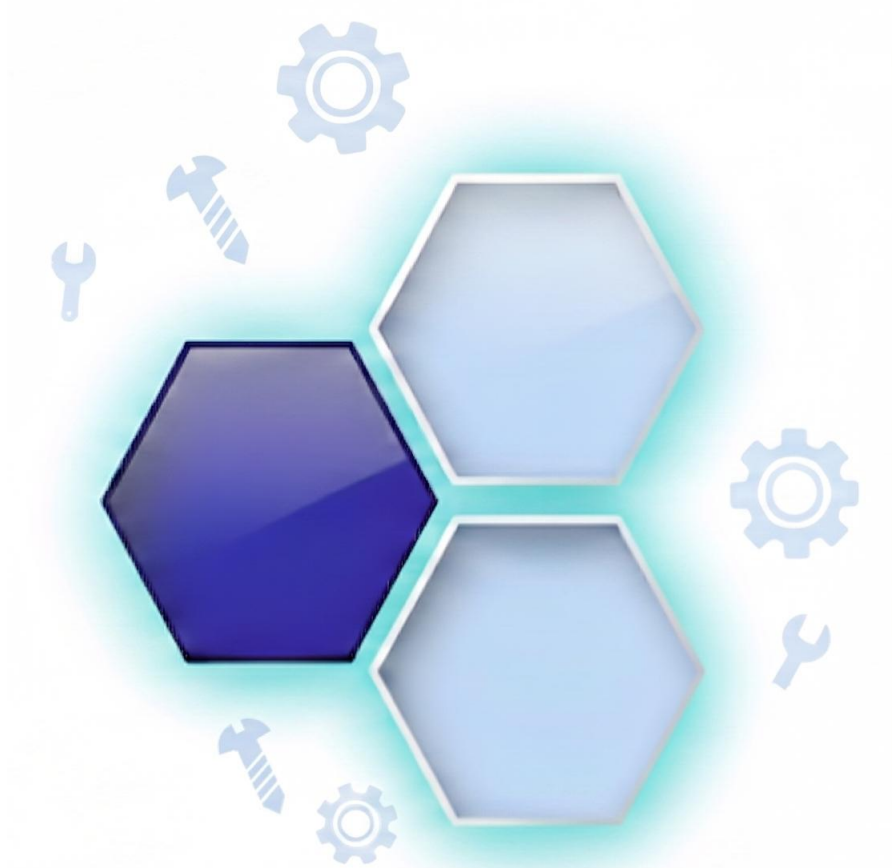
Automated ETL: extraction, transformation, and loading (ETL) from eMERGE 4 Clinical Data Platform REDCap to i2b2 using the **redcap2i2b2** R package

- Converts Research Electronic Data Capture (REDCap)** data dictionary and dataset into i2b2 **ontology** and **fact** structures
- Configuration files** enable flexible mapping across dates, categorical, and numeric fields aligned to i2b2 ontologies/metadata; designed for easy extensibility.
- We applied this to the **e4 Clinical Data Platform (CDP)** consisting of participant surveys and coordinator-entered EHR data

REDCap Field Type	Ontology representation	Data behavior
checkbox	☐ + (☐ x choices)	Boolean facts from column __n
dropdown / radio / yesno / truefalse	☐ + (☐ x choices)	Single column holds code; choose matching leaf
text / notes / calc / slider	☐ Single leaf	Store literal value (nval_num if numeric validation, else tval_char)

Electronic Health Record Outcomes Data

International Classification of Diseases (ICD) diagnoses, **RxNorm** medications, **Logical Observation Identifiers Names and Codes** labs, and **Current Procedural Terminology (CPT)** procedures.



- eMERGE IV Outcomes - Procedures (CPT4)
 - Anesthesia (1002796) (2014AA-2022AB)
 - Category II Codes (1014507) (2014AA-2022AB)
 - Category III Codes (1014405) (2014AA-2022AB)
 - Evaluation And Management Services (1013625) (2014AA-2022AB)
- eMERGE IV Outcomes - Medications (RXNORM)
 - Antihistamines (AH000)
 - Antimicrobials (AM000)
 - Antineoplastics (AN000)
 - Antiparasitics (AP000)
 - Autonomic Medications
 - Diagnostic Agents (DX000)
 - Gastrointestinal Medications (GA000)
- eMERGE IV Outcomes - Diagnoses (ICD10 - ICD9)
 - Certain Infectious and parasitic diseases (a00-b99)
 - Diseases of the circulatory system (i00-i99)
 - Diseases of the digestive system (k00-k94)
 - Diseases of the nervous system (g00-g99)
 - Diseases of the respiratory system (j00-j99)
- eMERGE IV Outcomes - Laboratory Tests (LOINC)
 - Blood Bank
 - Chemistry
 - Coagulation
 - Pathology
 - Serology
 - Urinalysis



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LLM-assisted Derived Data Derivation

- Observation Fact Generation:** Boolean-logic rules combine eMerge CDP variables, enabling production of **derived facts** (e.g., "patient meets high-risk criteria for breast cancer").
- Shared Clinical Ontology:** An expert-defined concept hierarchy enables intuitive query building.
- LLM-Assisted Data Conversion:** We leverage **Large Language Models (LLMs)** with comprehensive prompts detailing i2b2 data structures to generate Python conversion programs and accompanying unit tests, ensuring **technical correctness**. This practical approach drastically shortens the path from complex source **data** to usable **information**.



Atrial Fibrillation High PRS

[afib_prs_high_calc] = "1" AND
[afib_rr_header_calc] = "1" AND
[date_gira_ehr_upload_deid] <> ""
Breast Cancer Not High MG
[breast_cancer_rr_header_calc] = "0" AND
[date_gira_ehr_upload_deid] <> "" AND
[sex_at_birth_deid] = "Female" AND
[invitae_import_status] = "1"

Boolean logic rules for derived measures, defined by eMERGE researchers (top) and rules converted into an i2b2 hierarchy (bottom)

- eMERGE IV CDP Derived Facts
 - Participants with GIRA in EHR
 - Conditions
 - Asthma
 - Atrial Fibrillation
 - Participants assessed for atrial fibrillation not at high risk
 - Participants at high risk for atrial fibrillation
 - Participants with atrial fibrillation family history that triggered high risk
 - Participants with high atrial fibrillation PRS
 - Participants with LP/P finding in the LMNA gene associated with atrial fibrillation
- Breast Cancer
- Chronic Kidney Disease
- Colorectal Cancer
- Coronary Heart Disease
- Hypercholesterolemia
- Obesity

Overview of eMERGE survey data hierarchy (left) and REDCap dictionary conversion rules (right)